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标题	Solvent-Free Copper-Catalyzed Azide-Alkyne Cycloaddition under Mechanochemical Activation
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摘要（原文节选）

The ball-mill-based mechanochemical activation of metallic copper powder facilitates solvent-free alkyne-azide click reactions (CuAAC). All parameters that affect reaction rate (i.e., milling time, revolutions/min, size and milling ball number) have been optimized. This new, efficient, facile and eco-friendly procedure has been tested on a number of different substrates and in all cases afforded the corresponding 1,4-disubstituted 1,2,3-triazole derivatives in high yields and purities. The final compounds were isol ⋯

数据表 · Table 1（4 行，含数值行 3 行）

Screening of ball features.

Entry	Ball Number	Active Surface Area b (mm2)	Yield (%) a		
Small (Ø = 2 mm)	Medium (Ø = 5 mm)	Big (Ø = 10 mm)			
1	none	none	10	10,666	67
2	625	none	10	18,520	80
3	1500	48	none	30,144	99

脚注（逐字）：a: Isolated yield; b: Active surface area = surfaceballs + surfacejar.

正文对应句（逐字）：a: Isolated yield; b: Active surface area = surfaceballs + surfacejar.

数据表 · Table 2（4 行，含数值行 3 行）

Scale-up of mechanochemical CuAAC.

Entry	Octyl Azide (1a)	Yield a		
mmol	g	%	g	
1 b	1	0.1552	99	0.254
2 b	10	1.5524	98	2.519
3 c	50	7.7620	95	12.208

脚注（逐字）：a: Isolated yield; b: 1 eq Cu powder, 5 min, 650 rpm; 50 mL jar, 1500 small balls (Ø = 2 mm) and 48 medium balls (Ø = 5 mm); c: 1 eq Cu powder, 5 min, 650 rpm; 250 mL jar, 48 medium balls (Ø = 5 mm) and 10 very large balls (Ø = 30 mm).

正文对应句（逐字）：a: Isolated yield; b: Active surface area = surfaceballs + surfacejar.

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